COM661
Full Stack Strategies and Development

Coursework Clinic

Clinic Outline

1. Dataset Selection
2. Database queries and API endpoints
3. Implementation Issues
4. Testing and Documentation
5. The Submission Package

1. Dataset Selection

- Desirable features of a dataset
  - Structured as a JSON Array for importing into MongoDB
  - Sufficient documents to demonstrate pagination, searching, etc. (recommend 100 documents as a minimum)
  - Randomly-generated data is OK to enhance the dataset
  - A range of field types (text, numeric, boolean, hyperlinks - possibly to images)
  - At least one subdocument collection (such as reviews embedded within businesses)

2. Database Queries and API Endpoints

- Basic queries
  - CRUD (Create, Retrieve, Update, Delete) operations - essential to demonstrate at least one of each for documents and sub-documents
- Queries specific to the dataset
  - Fetching documents by specific field values rather than by _id
  - Queries on multiple fields
  - Range queries (with numeric fields)
  - Using the Aggregation Pipeline

3. Implementation Issues

- Code structure
  - Modular application using Blueprints OR well-structured and commented code that clearly identifies each section
- Authentication and User Management
- Endpoints to support functionality
  - e.g. to return the number of documents, number of pages of data, number of document with specific properties, etc.
- Flexible endpoints

4. Testing and Documentation

- Testing
  - “Success” testing – make sure that all endpoints provide the expected response for correct operation
  - “Failure” testing – make sure the appropriate HTTP Status code is returned for all error cases
- Documentation
  - Provide descriptions for all endpoints, parameters to API requests, and for the API as a whole
  - Publish documentation on the web

5. The Submission Package

- The application
  - All Python code files and all database collections as JSON files
- Supporting Evidence
  - Video (5 minutes max) split between demonstration and a code walkthrough
  - Link to Postman documentation
  - Screenshot of the result of running the complete collection of Postman tests
  - (Optional) Design and Implementation document for use of AI
- Upload to Blackboard with a Self-Evaluation Report